# ProblemSet03_Ex3.R

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```r
############################
# Load packages
############################
library(YieldCurve)
```

```
## Loading required package: xts

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
set.seed(13)

############################
# Prepare data
############################

yc.data <- read.csv2("YieldCurve.csv", header = TRUE, stringsAsFactors = FALSE)

# Extract chosen yield curve column
# Adjust column index if your YieldCurve.csv uses another column.
yields <- as.numeric(yc.data[, 2])

# ================================================================
# EXERCISE 1
# Nelson-Siegel-Svensson Model
# ================================================================
```

```r
# Exercise 1 a) Estimate NSS model
```

```r
maturity <- c(3/12, 6/12, 9/12, seq(1, 30, 1))

NSSmodel <- Svensson(yields, maturity)

# Svensson() returns tau1 and tau2, which are reciprocals of lambda1/lambda2
```

```r
b0 <- NSSmodel[1]
b1 <- NSSmodel[2]
b2 <- NSSmodel[3]
b3 <- NSSmodel[4]

lambda1 <- 1 / NSSmodel[5]
lambda2 <- 1 / NSSmodel[6]

cat("\nNSS parameter estimates\n")
```

```
##
## NSS parameter estimates
```

```r
cat(sprintf("b0      : %.6f\n", b0))
```

```
## b0      : 1.497137
```

```r
cat(sprintf("b1      : %.6f\n", b1))
```

```
## b1      : 0.423610
```

```r
cat(sprintf("b2      : %.6f\n", b2))
```

```
## b2      : 1.651342
```

```r
cat(sprintf("b3      : %.6f\n", b3))
```

```
## b3      : 6.602684
```

```r
cat(sprintf("lambda1 : %.6f\n", lambda1))
```

```
## lambda1 : 1.434619
```

```r
cat(sprintf("lambda2 : %.6f\n", lambda2))
```

```
## lambda2 : 0.065210
```

```r
# Spot yield curve function y0(T)
```

```r
# y0(T) returns DECIMAL rates, e.g. 0.025 for 2.5%.
y0 <- function(T){
  F1 <- (1 - exp(-lambda1 * T)) / (lambda1 * T)
  F2 <- (1 - exp(-lambda1 * T)) / (lambda1 * T) - exp(-lambda1 * T)
  F3 <- (1 - exp(-lambda2 * T)) / (lambda2 * T) - exp(-lambda2 * T)

  (b0 + b1 * F1 + b2 * F2 + b3 * F3) / 100
}
```

```r
# Exercise 1 c) Arbitrage-free forward rate function f0(T1,T2)
```

```r
# f0(T1,T2) returns DECIMAL rates.
f0 <- function(T1, T2){
  (y0(T2) * T2 - y0(T1) * T1) / (T2 - T1)
}

# T_1 <- 1
# T_2 <- seq(T_1 + 0.01, 30, by = 0.01)
# f0_curve <- f0(T_1, T_2)
```

```r
# Exercise 1 d) Instantaneous forward rate function f0inst(T)
```

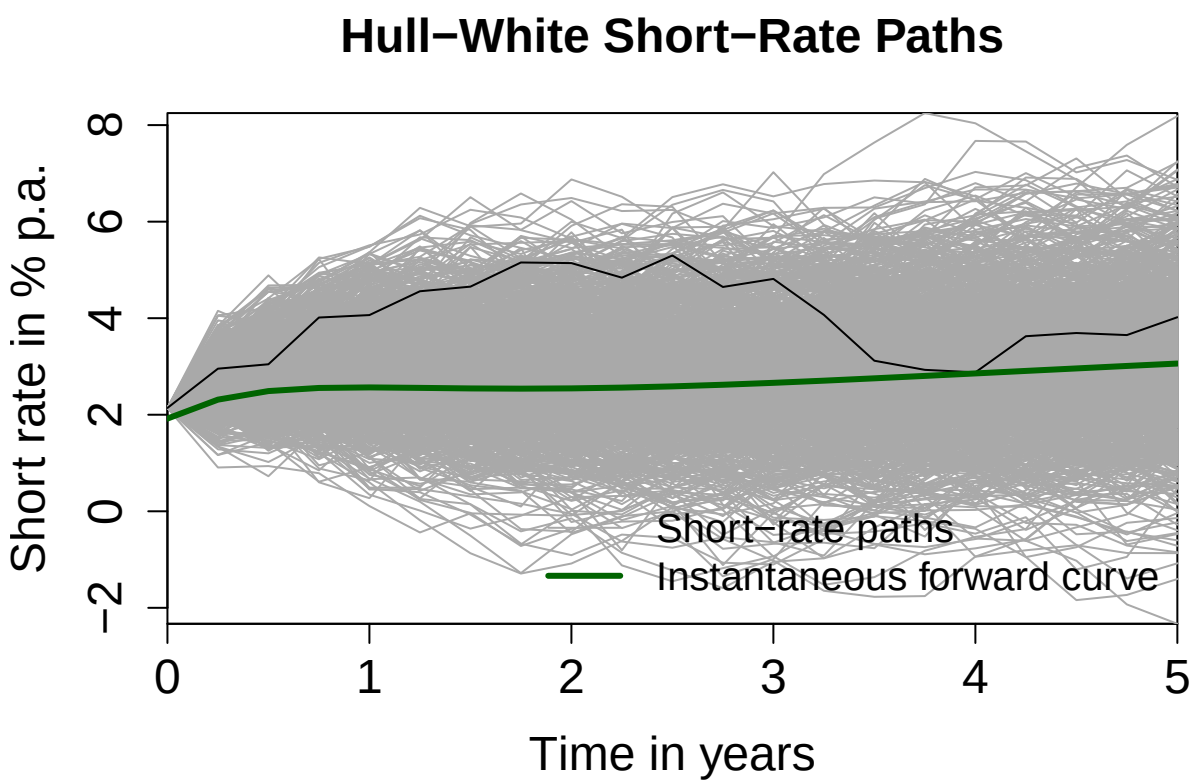
```r
# f0inst(T) returns DECIMAL rates.
f0inst <- function(T){
  F1 <- exp(-lambda1 * T)
  F2 <- exp(-lambda1 * T) * lambda1 * T
  F3 <- exp(-lambda2 * T) * lambda2 * T

  (b0 + b1 * F1 + b2 * F2 + b3 * F3) / 100
}

# T_grid <- seq(0.25, 30, by = 0.01)
# f0inst_curve <- f0inst(T_grid)

# ================================================================
# EXERCISE 2
# Hull-White Model
# ================================================================
```

```r
# Simulate Hull-White short-rate paths
```

```r
paths <- 1000
alpha <- 0.20
sigma <- 0.01
T <- 5              # Maturity T

# quarterly increments for the MC simulation
s <- 4

dt <- rep(1 / s, T * s)
t <- c(0, cumsum(dt))

# Initial short rate: very short maturity spot rate
r0 <- y0(1 / s)

# Derivative of instantaneous forward curve
df0inst_dt <- function(T){
  (
    -lambda1 * exp(-lambda1 * T) * (b1 - b2 + b2 * lambda1 * T) +
      lambda2 * exp(-lambda2 * T) * (b3 - b3 * lambda2 * T)
  ) / 100
```

```r
}

# Simulate short-rate paths -- LOOP version
rt <- matrix(NA, nrow = paths, ncol = length(t))
rt[, 1] <- r0

for(j in 1:paths){
  for(i in 2:length(t)){

    # Time-varying mean-reversion level
    theta <- df0inst_dt(t[i - 1]) + alpha * f0inst(t[i - 1])

    # Wiener increment: detla_W ~ N(0, delta_t)
    dW <- rnorm(n = 1, mean = 0, sd = sqrt(dt[1]))

    # Euler discretization of Hull-White:
    # r_t = r_{t-dt} + (theta_t - alpha*r_{t-dt})*dt + sigma*dW
    rt[j, i] <- rt[j, i - 1] +
      (theta - alpha * rt[j, i - 1]) * dt[1] +
      sigma * dW
  }
}

# Plot simulated short-rate paths with instantaneous forward curve
short_rate_ylim <- range(c(rt, f0inst(t)), na.rm = TRUE) * 100

plot(t, rt[1, ] * 100,
     type = "l",
     lwd = 1,
     xlim = range(t),
     ylim = short_rate_ylim,
     xaxs = "i",
     yaxs = "i",
     col = "darkgrey",
     cex.axis = 1.5,
     cex.lab = 1.5,
     cex.main = 1.5,
     main = "Hull-White Short-Rate Paths",
     xlab = "Time in years",
     ylab = "Short rate in % p.a.")

for(j in 2:paths){
  lines(t, rt[j, ] * 100,
        lwd = 1,
        col = "darkgrey")
}

lines(t, rt[1, ] * 100,
      lwd = 1,
      col = "black")

lines(t, f0inst(t) * 100,
      lwd = 3,
```

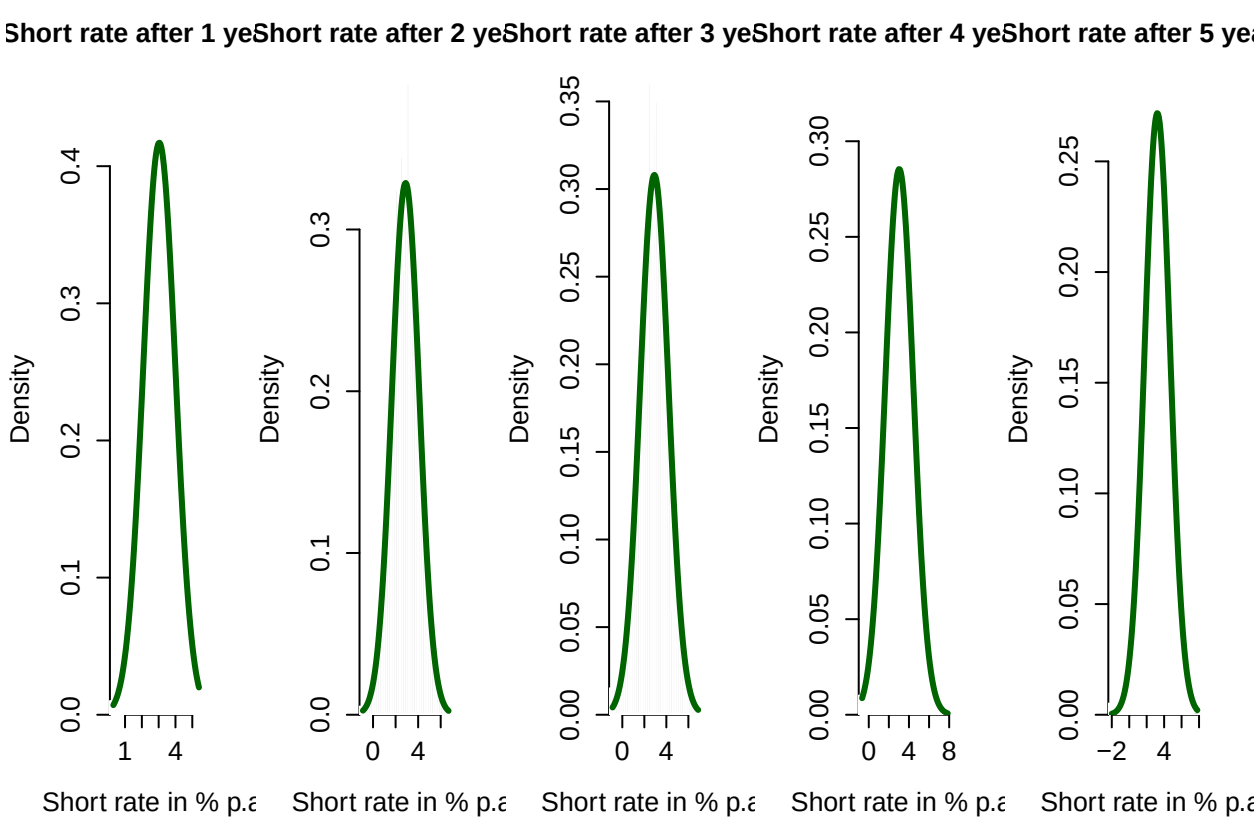
```
      col = "darkgreen")

legend("bottomright",
       legend = c("Short-rate paths", "Instantaneous forward curve"),
       lty = 1,
       lwd = c(1, 3),
       col = c("darkgrey", "darkgreen"),
       bty = "n",
       cex = 1.25)
```




```
# Histograms after 1 to 5 years
```

```
years <- c(seq(1, T))

year.idx <- years * s + 1
rt.selected <- rt[, year.idx]

colnames(rt.selected) <- paste0("t = ", years)

par.old <- par(no.readonly = TRUE)
par(mfrow = c(1, 5))

for(i in seq_along(years)){

  rt_i <- rt.selected[, i]
```

```r
  mu_i <- mean(rt_i)
  sd_i <- sd(rt_i)

  hist(rt_i * 100,
       breaks = 40,
       probability = TRUE,
       xaxs = "i",
       yaxs = "i",
       col = "lightgrey",
       border = "white",
       main = paste("Short rate after", years[i], "year(s)"),
       xlab = "Short rate in % p.a.",
       ylab = "Density",
       cex.axis = 1.2,
       cex.lab = 1.2,
       cex.main = 1.2)

  curve(dnorm(x, mean = mu_i * 100, sd = sd_i * 100),
        add = TRUE,
        col = "darkgreen",
        lwd = 3)
}
```



```r
par(par.old)

rt.summary <- data.frame(
  year = years,
  mean_decimal = colMeans(rt.selected),
  sd_decimal = apply(rt.selected, 2, sd),
  mean_percent = colMeans(rt.selected) * 100,
  sd_percent = apply(rt.selected, 2, sd) * 100
)

cat("\nShort-rate summary\n")
```

```
##
## Short-rate summary
```

```r
print(rt.summary)
```

```
##       year mean_decimal  sd_decimal mean_percent sd_percent
## t = 1    1   0.03037212 0.009563471     3.037212  0.9563471
## t = 2    2   0.02886390 0.012133353     2.886390  1.2133353
## t = 3    3   0.02909509 0.012948168     2.909509  1.2948168
## t = 4    4   0.03034108 0.013973072     3.034108  1.3973072
## t = 5    5   0.03223674 0.014676149     3.223674  1.4676149
```

```r
# Expected future zero-coupon bond price
```

```r
# Initial zero-coupon bond price from today's curve
P0 <- function(T){
  exp(-y0(T) * T)
}

# Hull-White B(t,T)
B <- function(t_bond, T_bond){
  (1 - exp(-alpha * (T_bond - t_bond))) / alpha
}

# Hull-White ln A(t,T)
ln_A <- function(t_bond, T_bond){
  log(P0(T_bond) / P0(t_bond)) +
    B(t_bond, T_bond) * f0inst(t_bond) -
    sigma^2 / (4 * alpha) *
    (1 - exp(-2 * alpha * t_bond)) *
    B(t_bond, T_bond)^2
}

# Hull-White log zero-coupon bond price
ln_P_tilde_t <- function(t_bond, T_bond, r_t){
  ln_A(t_bond, T_bond) -
    B(t_bond, T_bond) * r_t
}
```

```r
# Pricing of the Short-Rate Realized Volatility Bonus Product:
# Done under traditional risk-neutral measure Q*
```

```r
# REMARK: T and s set above since already needed for the simulation
#         of the Hull-White short-rate paths

# self defined assumptions/definitions
notional <- 10000000
K_vol = 0.0075                # volatility strike = 0.75%

# Index of payoff date
idx_pay <- T * s + 1   # with s = 4, this gives 21

# Short-rate paths from time 0 to product maturity T
rt_product <- rt[, 1:idx_pay]

# Quarterly short-rate changes:
# delta_r[j, i] = r_j(t_i) - r_j(t_{i-1})
delta_r <- rt_product[, -1] - rt_product[, -ncol(rt_product)]

# Short-rate paths (= 1000 ; set above) from time 0 to time T_product
rt_product <- rt[, 1:idx_pay]

# Realized annualized short-rate volatility per path
RV_T <- sqrt(rowSums(delta_r^2) / T)

# Payoff at T_product
payoff_T <- notional * pmax(RV_T - K_vol, 0)

# Pathwise discount factor from 0 to T_product under Q*
DF_0_T <- exp(
  -rowSums(rt_product[, 1:(ncol(rt_product) - 1), drop = FALSE]) * dt[1]
)

# Monte Carlo price
volatility_bonus_price <- mean(DF_0_T * payoff_T)

# Monte Carlo standard error
volatility_bonus_price_se <- sd(DF_0_T * payoff_T) / sqrt(paths)

cat("\nShort-Rate Realized Volatility Bonus Pricing under Q*\n")
```

```
##
## Short-Rate Realized Volatility Bonus Pricing under Q*
```

```r
cat(sprintf("Expected realized volatility: %.6f\n", mean(RV_T)))
```

```
## Expected realized volatility: 0.010373
```

```r
cat(sprintf("Price today: %.6f\n", volatility_bonus_price))
```

```
## Price today: 24858.770339
```

```r
cat(sprintf("Monte Carlo SE: %.6f\n", volatility_bonus_price_se))
```

```
## Monte Carlo SE: 422.704705
```